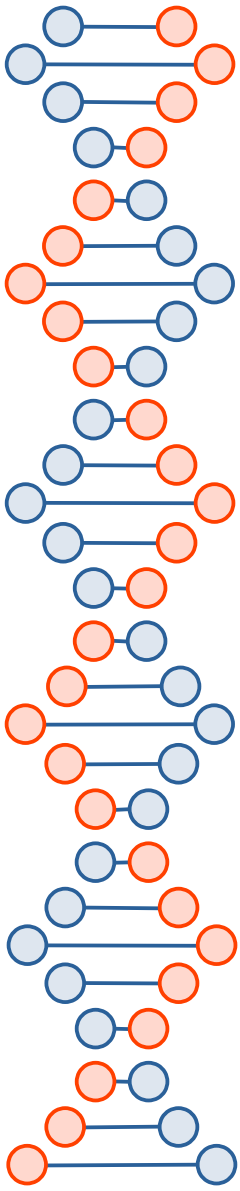


The Bias-Variance Trade-Off

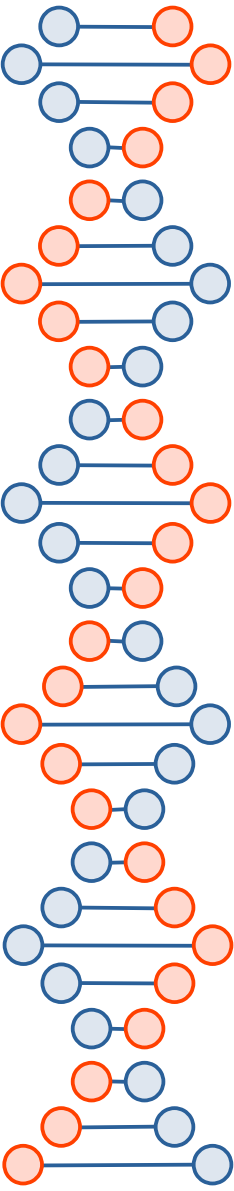
Dr. Josafá Pontes

Machine Learning



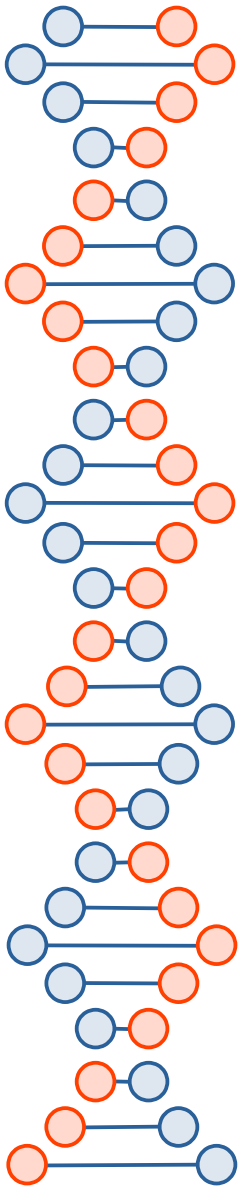
Introduction

- All learning error can be broken down into bias or variance error.
- Bias refers to the simplifying assumptions made by the algorithm to make the problem easier to solve.
- Variance refers to the sensitivity of a model to changes to the training data.
- Bias and variance framework helps to understand predictions from applied machine learning.



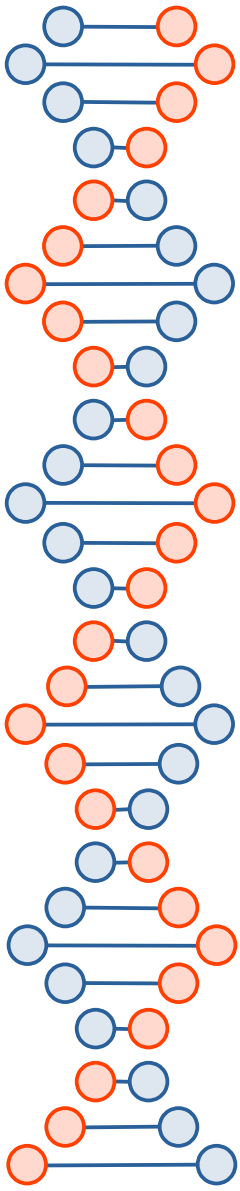
Overview of Bias and Variance

- In supervised machine learning an algorithm learns a model from training data. The goal of
- Any supervised machine learning algorithm is to best estimate the mapping function (f) for the output variable (Y) given the input data (X).
- The mapping function is often called the target function because it is the function that a given supervised machine learning algorithm aims to approximate.
- The prediction error for any machine learning algorithm can be broken down into three parts:
 - Bias Error
 - Variance Error
 - Irreducible Error



Types of Errors

- The irreducible error cannot be reduced regardless of what algorithm is used.
- It is the error introduced from the chosen framing of the problem and may be caused by factors like noise and overlapping variables that influence the mapping from the input variables to the output variable.
- Our focus is on:
 - bias error
 - variance error

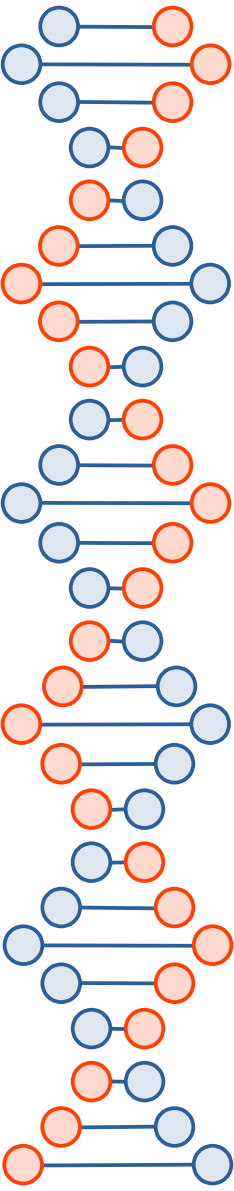


The Bias Error

- Bias are the simplifying assumptions made by a model to make the target function easier to learn.
- Generally parametric algorithms have a high bias making them fast to learn and easier to understand but generally less flexible.
- In turn they have lower predictive performance on complex problems that fail to meet the simplifying assumptions of the algorithms bias.
 - Low Bias: Suggests less assumptions about the form of the target function.
 - High-Bias: Suggests more assumptions about the form of the target function.
- Examples of **low-bias** machine learning algorithms include: Decision Trees, k-Nearest Neighbors and Support Vector Machines.
- Examples of **high-bias** machine learning algorithms include: Linear Regression, Linear Discriminant Analysis and Logistic Regression.

The Variance Error (1)

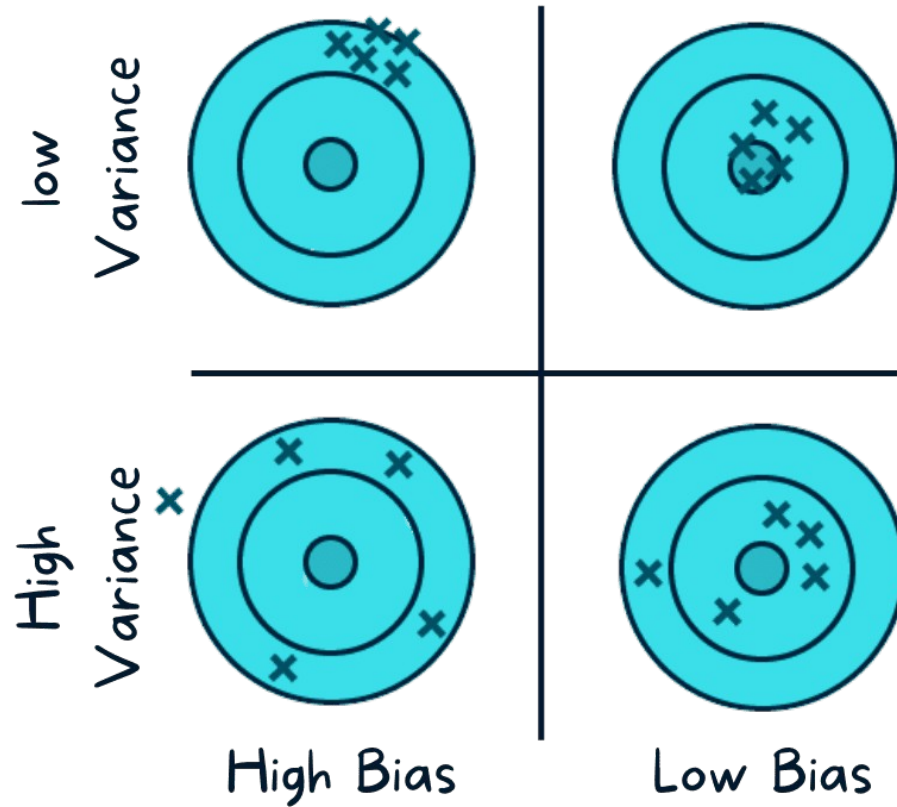
- Variance is the amount that the estimate of the target function will change if different training data was used.
- The target function is estimated from the training data by a machine learning algorithm, so we should expect the algorithm to have some variance.
- Ideally, it should not change too much from one training dataset to the next, meaning that the algorithm is good at picking out the hidden underlying mapping between the inputs and the output variables.
- Machine learning algorithms that have a high variance are strongly influenced by the specifics of the training data.
- This means that the specifics of the training data influences the number and types of parameters used to characterize the mapping function.

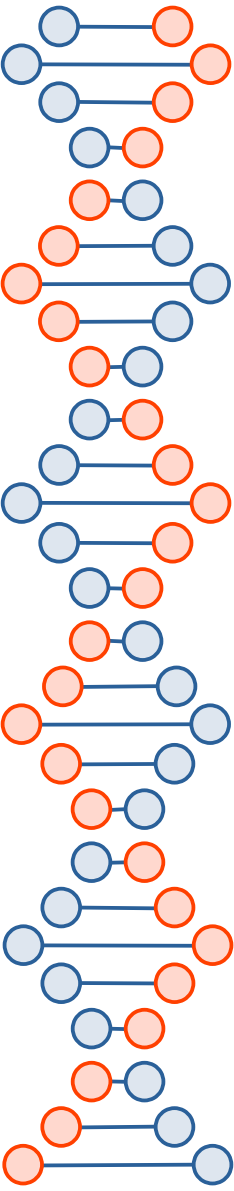


The Variance Error (2)

- Low Variance: Suggests small changes to the estimate of the target function with changes to the training dataset.
- High Variance: Suggests large changes to the estimate of the target function with changes to the training dataset.
- Generally nonparametric machine learning algorithms that have a lot of flexibility have a high variance.
- For example decision trees have a high variance, that is even higher if the trees are not pruned before use.
- Examples of **low-variance** machine learning algorithms include: Linear Regression, Linear Discriminant Analysis and Logistic Regression.
- Examples of **high-variance** machine learning algorithms include: Decision Trees, k-Nearest Neighbors and Support Vector Machines.

The Bias and Variance Errors





Bias-Variance Trade-Off (1)

- The goal of any supervised machine learning algorithm is to achieve low bias and low variance.
- In turn the algorithm should achieve good prediction performance.
- You can see a general trend in the examples above:
 - Parametric or linear machine learning algorithms often have a high bias but a low variance.
 - Nonparametric or nonlinear machine learning algorithms often have a low bias but a high variance.

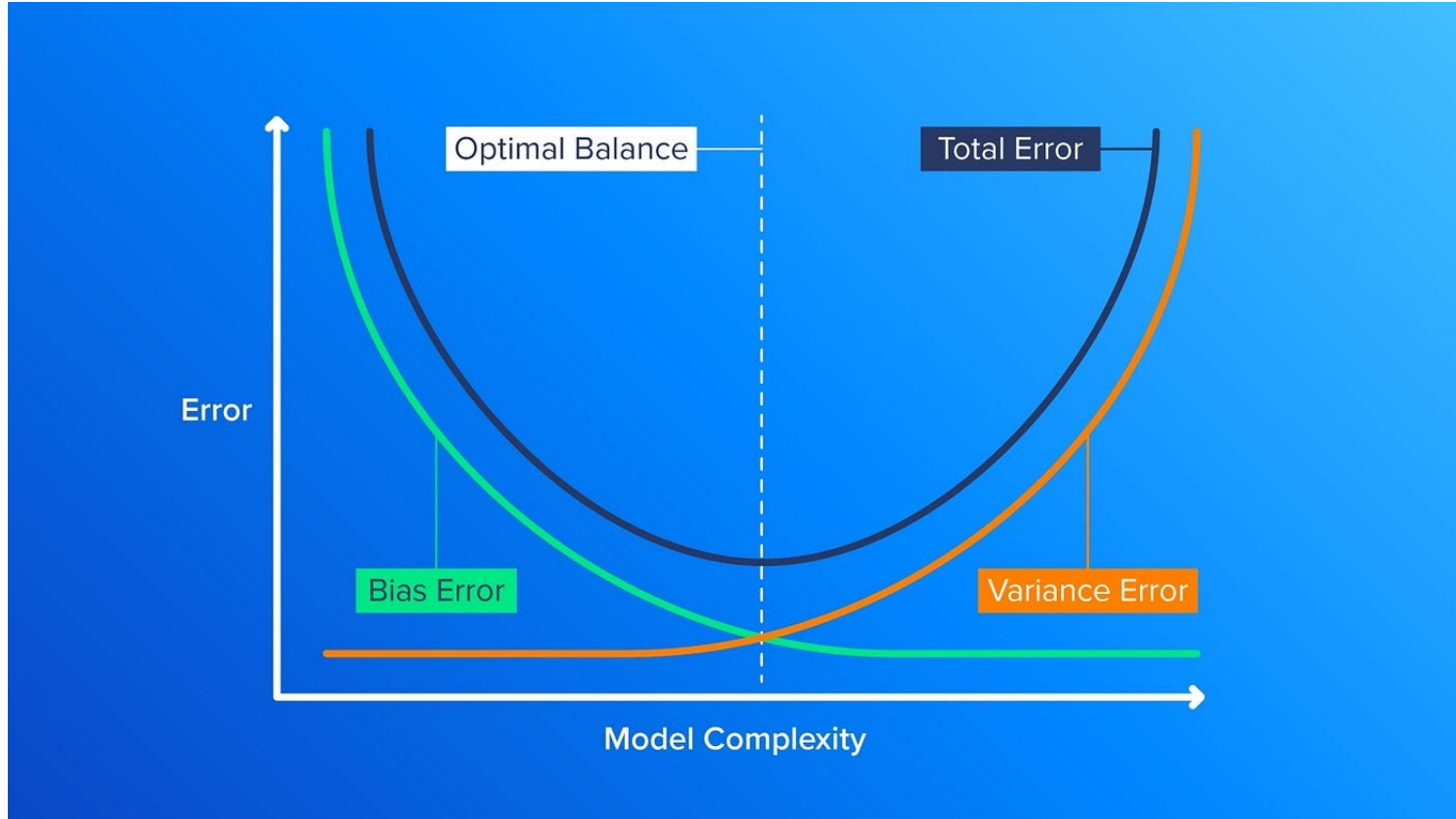
Bias-Variance Trade-Off (2)

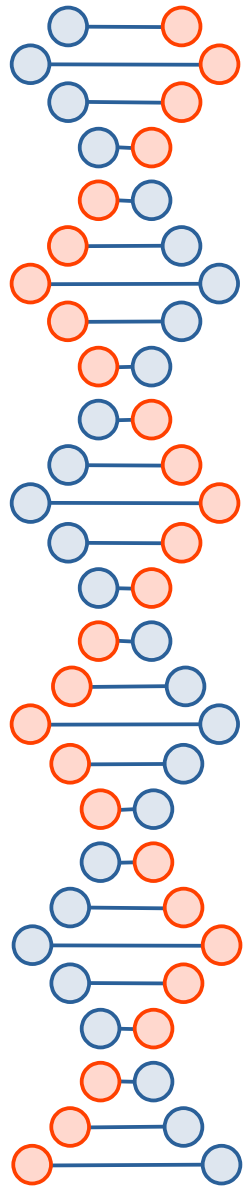
- The parameterization of machine learning algorithms is often a battle to balance out bias and variance.
- Below are two examples of configuring the bias-variance trade-off for specific algorithms:
 - The k-nearest neighbors algorithm has low bias and high variance, but the trade-off can be changed by increasing the value of k which increases the number of neighbors that contribute to the prediction and in turn increases the bias of the model.
 - The support vector machine algorithm has low bias and high variance, but the trade-off can be changed by increasing the C parameter that influences the number of violations of the margin allowed in the training data which increases the bias but decreases the variance.

Bias-Variance Trade-Off (3)

- There is no escaping the relationship between bias and variance in machine learning.
 - Increasing the bias will decrease the variance.
 - Increasing the variance will decrease the bias.
- There is a trade-off at play between these two concerns and the algorithms you choose and the way you choose to configure them are finding different balances in this trade-off for your problem.
- In reality we cannot calculate the real bias and variance error terms because we do not know the actual underlying target function.
- Nevertheless, as a framework, bias and variance provide the tools to understand the behavior of machine learning algorithms in the pursuit of predictive performance.

Bias-Variance Trade-Off (4)





Summary (1)

Cheat Sheet – Bias-Variance Tradeoff

What is Bias?

$$\text{bias} = \mathbb{E}[f'(x)] - f(x)$$

- Error between average model prediction and ground truth
- The bias of the estimated function tells us the capacity of the underlying model to predict the values

What is Variance?

$$\text{variance} = \mathbb{E}[(f'(x) - \mathbb{E}[f'(x)])^2]$$

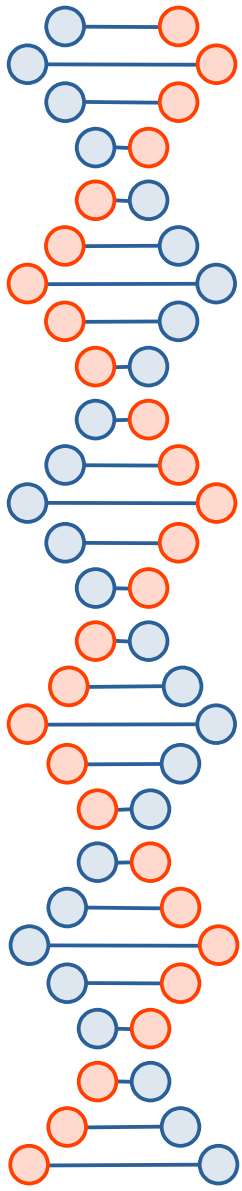
- Average variability in the model prediction for the given dataset
- The variance of the estimated function tells you how much the function can adjust to the change in the dataset

High Bias

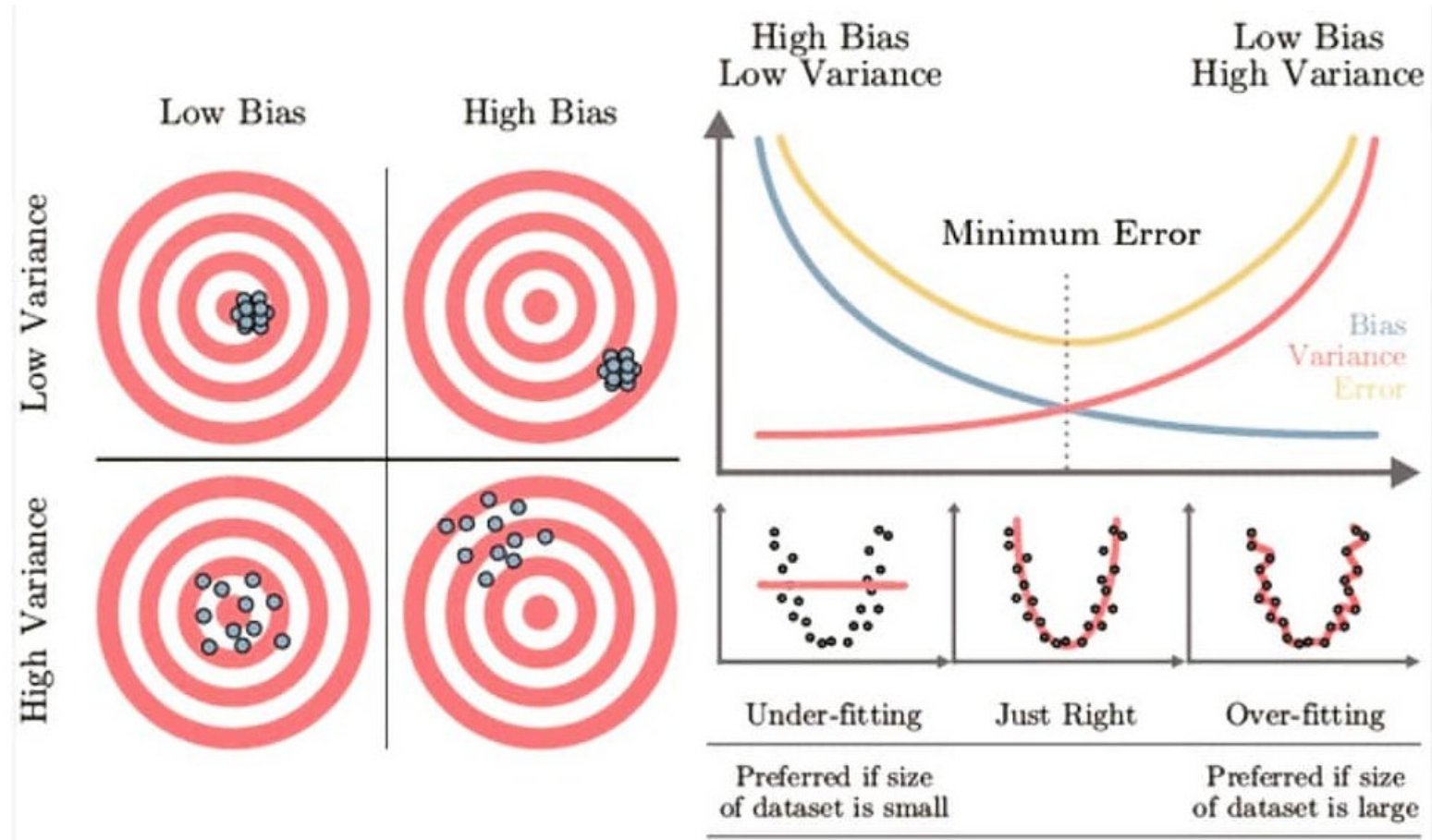
- Overly-simplified Model
- Under-fitting
- High error on both test and train data

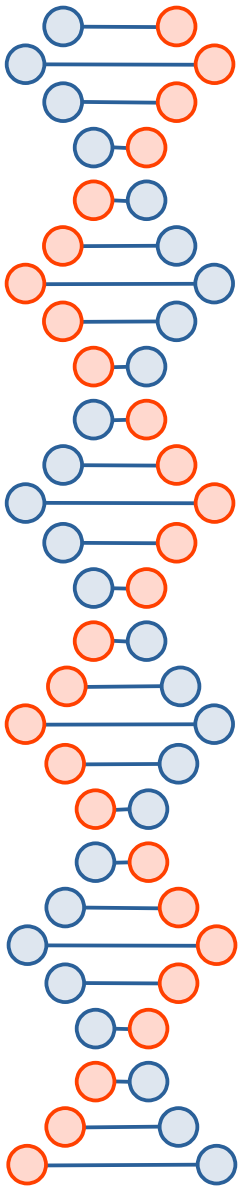
High Variance

- Overly-complex Model
- Over-fitting
- Low error on train data and high on test
- Starts modelling the noise in the input



Summary (2)





Summary (3)

Bias variance Trade-off

- Increasing bias reduces variance and vice-versa
- $\text{Error} = \text{bias}^2 + \text{variance} + \text{irreducible error}$
- The best model is where the error is reduced.
- Compromise between bias and variance